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FEDERAL SPECIFICATION

GROOVERS, EDGERS, AND CORNER TOOLS - CONCRETE, HAND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers groovers, edgers, and corner tools for grooving joints in pavements, edging pavements and curbing, and shaping steps and corners.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The tools covered by this specification shall be of the following types and classes, as specified (see 6.1):

Type I - Groover
Type II - Edger
Type III - Corner tool
  Class 1 - Inside
  Class 2 - Outside

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

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(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

D 4101 - Standard Specification for Propylene Elastic Injection and Extrusion Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of groovers, edgers, or corner tools otherwise conforming to this specification.

3.2 Material. Material used in construction of the tools shall be as specified. All forgings shall be of uniform quality and free from all defects and imperfections that might affect serviceability or durability.

3.3 Design. Groovers, edgers, and corner tools shall consist of a blade and handle. The design shall preclude the possibility of the assembled parts becoming loose or the handle turning when used under normal service conditions.

3.3.1 Blades. Blades for types I and II tools shall be made of bronze or stainless steel and shall have either both ends turned up or one end straight and one end turned up. Blades for type III, corner tools, shall be of bronze or stainless steel. When steel is used for blades, it shall be 0.055 ± 0.010 inch thick. The face, or working surface, of types I and II blades shall be ground and highly polished. The blade of type III, corner tools, shall have a smooth finish. When placed on a flat surface, blades of types II and III shall show no evidence of warpage.

3.3.2 Handles.

3.3.2.1 Wood. Wood handles shall be of clear, straight-grained hardwood. Handles shall be smoothly finished, free from cracks or splinters, and coated with shellac, varnish, or other equally suitable transparent substance to prevent moisture absorption, or shall have a natural finish.

3.3.2.2 Plastic. Plastic handles shall be of high-impact strength in accordance with ASTM D 4101, and shall withstand the tests specified in 4.4.

3.4 Type I, groover. The type I groover shall have a groove cutter midway between the sides of the blade. The groover shall conform to table I, and be similar to figure 1.

TABLE I. Type I, groover

Blade				Groove cutter				Handle			
Length		Width		Depth		Width		Diameter		Length	
Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inch	Inch	Inch	Inch	Inches	Inches	Inches
5-3/4	6-1/4	2-5/8	3	3/8	5/8	3/8	5/8	1	1-3/8	3-3/4	4-1/2
8-3/4	9-1/4	2-1/2	3	3/8	5/8	3/8	5/8	1	1-3/8	3-3/4	4-1/2

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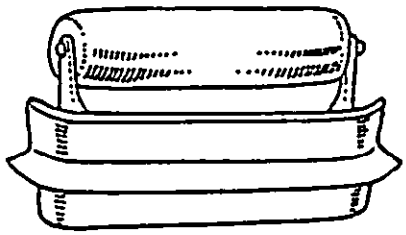


Figure 1. Type I, groover.

3.5 Type II, edger. The type II edger shall have one side turned down into an underlip 90° from the working surface, and shall have the ends of the underlip turned out. The edgers shall conform to table II, and be similar to figure 2.

TABLE II. Type II, edger

Blade				Underlip				Handle			
Length		Width		Depth		Radius		Diameter		Length	
Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inch	Inch	Inch	Inches	Inches	Inches	Inches
5-3/4	6-1/4	2-5/8	3	9/16	11/16	1/4	13/32	1-1/16	1-3/8	3-3/4	4-3/8

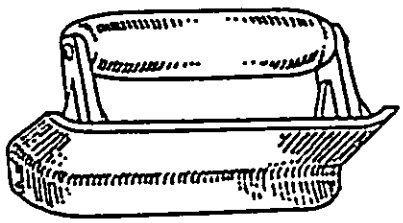


Figure 2. Type II, edger.

3.6 Type III, corner tool.

3.6.1 Class 1, inside. The working surfaces of the class 1 inside corner tool shall be at an angle of 90° plus 0° minus 1° to each other. The tool shall conform to table III, and be similar to figure 3.

TABLE III. Type III, class 1 and 2, inside and outside, corner tool

Blade				Radius ± 1/16	Handle			
Length		Width each side			Diameter		Length	
Min.	Max.	Min.	Max.		Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inches	Inches	Inches	Inches
5-7/8	6-1/8	2-7/16	2-9/16	1/4	1-7/32	1-7/16	4-3/8	4-5/8

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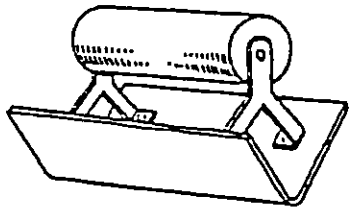


Figure 3. Type III, class 1 inside corner of tool, concrete.

3.6.2 Class 2, outside. The working surfaces of the outside corner tool shall be at an angle of 90° plus 0° minus 1° to each other. The tool shall conform to table III, and be similar to figure 4.

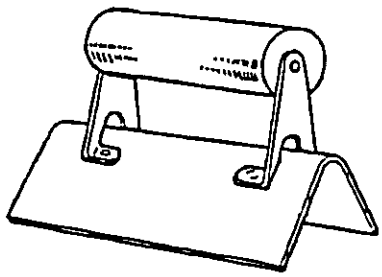


Figure 4. Type III, class 2 outside corner tool, concrete.

3.7 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.8 Workmanship. Workmanship shall be of a quality that precludes defects which could affect the appearance, serviceability, or durability of the tool.

3.9 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Inspection.

4.3.1 Visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in terms of groovers, edgers, or corner tools. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for total defects.

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TABLE IV. Classification of defects

Examine	Defect
Construction and Workmanship	Component missing or improperly assembled
	Not specified dimensions
	Blade and handle loose
	Groove cutter missing or not as specified (type I)
	Underlip missing or of incorrect dimensions (type II)
	Angle of working surface not within tolerance (type III)
Handle	Turns under normal service condition
	Wood cracked or splintered
Blades	Thickness not within specified tolerance
	Not turned up (types I and II)
	Warped (types II and III)
Material	Not as specified
	Wood not clear, straight-grained hardwood
	Plastic handles not high-impact strength
Finish	Handles not smoothly finished, as specified
	Working surface not ground and highly polished (types I and II)
	Not smooth (type III)
Marking	Missing, incomplete, not legible
	Not permanent

4.3.2 Dimensional examination. The end items shall be examined for any nonconformance with dimensional requirements, as specified in 3.3.1, 3.4, 3.5, and 3.6, and listed in tables I, II, and III. The lot size shall be expressed in terms of groovers, edgers, or corner tools. The sample unit shall be one groover, edger, or corner tool. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4 Testing. The end items shall be tested for the characteristics specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3. The lot size shall be expressed in units of groovers, edgers, or corner tools. The sample unit shall be one tool. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.1 Plastic handle tests.

4.4.1.1 Impact test. The handle, with a suitable guard around it, shall be placed on a solid base in a manner to permit dropping a weight directly on the handle. The handle shall not break, crack, or distort when a 5-pound weight has been dropped three times as specified in table V.

TABLE V. Impact test

Height of drop weight	
At room temperature	At minus 20°F
Feet	Feet
5	3

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4.4.1.2 Flammability test. The blade of the tool shall be clamped in a support with the handle extended outward in a horizontal position. An approximately 5 inch square of 10-mesh Bunsen burner gauze shall be clamped in a horizontal position 1/4-inch below the bottom of the handle, with 1/2-inch of the handle extending beyond the edge of the gauze. A Bunsen burner with an efficient flame 1/2-inch to 3/4-inch in height shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handle after the removal of the Bunsen burner. In case the handle does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinguishment of the flame. If the handle does not continue to burn until the flame has reached the blade end of the handle after the second ignition, the sample shall be considered self-extinguishing. If the sample continues to burn, the burning rate shall not exceed 2 inches per minute.

4.4.1.3 Heat distortion. The handle of the tool being tested shall be completely immersed in actively boiling water for a period of 2 minutes. At the end of this period, the sample shall be removed from the water and shall show no significant signs of distortion or blistering.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VI. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE VI. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling Distortion of container
Contents	Number per container is more or less than required Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Level of preservation, packaging, and packing required (see 5.1).
d. Marking, if other than specified.

MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an official interest in this Federal Specification

PREPARING ACTIVITY:

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